

Lab 5: The average treatment effect

Causal Machine Learning

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We will be using data from an observational study of 996 patients receiving a Percutaneous Coronary Intervention (PCI) with/without augmented therapy with abciximab (a IIb/IIIa cascade blocker) at Ohio Heart Health in 1997 and the patients are followed for at least 6 months by the staff of the Lindner Center. The outcome of interest is whether they survived at least 6 months. Note that patients receiving abciximab tended to be more severely diseased than those who did not.

Read the package `twang` in R and the data set `lindner` as follows.

```
> library(twang)
> data(lindner)
```

To evaluate the effect of augmented PCI (the information is held in the variable `abcix`) on 6 months survival, we create a binary indicator of whether patients have died within 6-months:

```
> lindner$survival_binary<-ifelse(lindner$sixMonthSurvive==FALSE,1,0)
> attach(lindner)
```

1. Carry out a naive (unadjusted) analysis to estimate the marginal association between treatment and outcome in terms of a risk difference (RD) and/or odds ratio (OR).
2. Use G-computation to estimate the average treatment effect (ATE). Try in the following ways:
 - (a) Estimate the ATE using a parametric model for the nuisance parameters. Interpret your results. Note how much simpler interpretation is relative to question 1.
 - (b) You can use the R package `stdReg` to obtain the standard error. You can alternatively use the functions `estimate_mu_par`, `est_gcomp` and `bootstrap_se_gcomp`.
 - (c) What are the advantages and disadvantages of using parametric models for G-computation?
 - (d) Estimate the ATE, now using the R package `SuperLearner` to estimate nonparametrically the nuisance parameters (you can do it without cross fitting).
 - (e) What are the advantages and disadvantages of using nonparametric machine learning algorithms for G-computation?
 - (f) Do you see ways to obtain an accompanying standard error?
3. Use Inverse Probability Weighting (IPW) to estimate the average treatment effect. Try in the following ways:
 - (a) Estimate the ATE using a parametric model for the propensity scores. Note that you can also use a weighted linear regression for this, or make use of the functions `estimate_pi_par` and `est_ipw`.
 - (b) Estimate the ATE, now using the R package `SuperLearner` to estimate nonparametrically the propensity scores (you can do it without cross fitting) and then a weighted linear regression. Note that you can also use the functions `estimate_pi_nonpar` and `est_ipw` for this.

- (c) Check the distribution of the estimated propensity scores as well as the inverse probability weights for the 2 treatment groups. Is there evidence of positivity violations?
- 4. Use Augmented Inverse Probability Weighting (AIPW) to estimate the average treatment effect. Try in the following ways:
 - (a) Estimate the ATE using the outcome model and propensity score estimates obtained in the previous steps.
 - (b) Compute the standard error based on the ATE's efficient influence curve.
 - (c) Now estimate the ATE using the R package `npcausal` or the function `est_aipw`. Change the number of folds (e.g., 1, 2, 5) and evaluate how that impacts results.
 - (d) Which are the advantages of AIPW with relative G-computation and IPW? Do you see any drawback?
- 5. Use Targeted Maximum Likelihood to estimate the average treatment effect. Try in the following ways:

- (a) Estimate the ATE using **SuperLearner** to estimate the nuisance parameters (you can do it without cross-fitting) and using the fluctuation model of the form

$$\text{logit } E(Y \mid A, L) = \text{logit } \bar{Q}_n^{(0)}(L) + \delta \frac{A}{g_n(L)}$$

- (b) Compute the standard error and 95% confidence intervals?
- (c) Repeat points (a) and (b) using the fluctuation model of the form

$$\text{logit } E(Y \mid A, L) = \text{logit } \bar{Q}_n^{(0)}(L) + \delta$$

- (d) Estimate the ATE, now using the R package `tmle`. Evaluate if you find different results with a more extensive library of learners to use in the `SL.library` argument of the **SuperLearner** package. Note that you can alternatively use the functions `estimate_mu_nonpar` (you can use the outcome model estimates from the previous steps) and `est_tmle`.
- (bonus) Repeat question (a), now implementing cross-fitting yourself. Think carefully if you should use the same fold to estimate both the nuisance parameters and the fluctuation parameter δ .

- (bonus) If you have time left, you can repeat question 4 and 5, now focussing on the average treatment effect in the treated.